

Optimizing US English vowels for word-level perception predicts the *cot-caught* merger

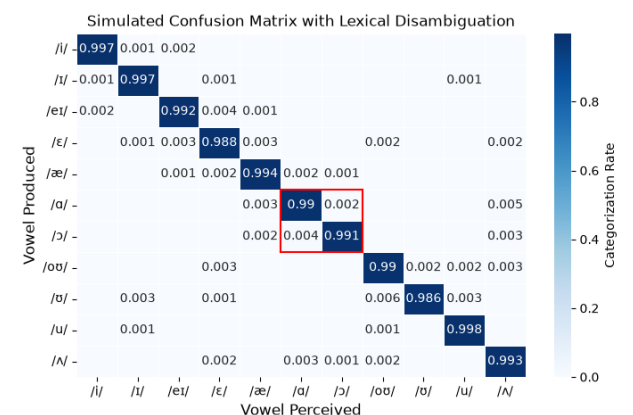
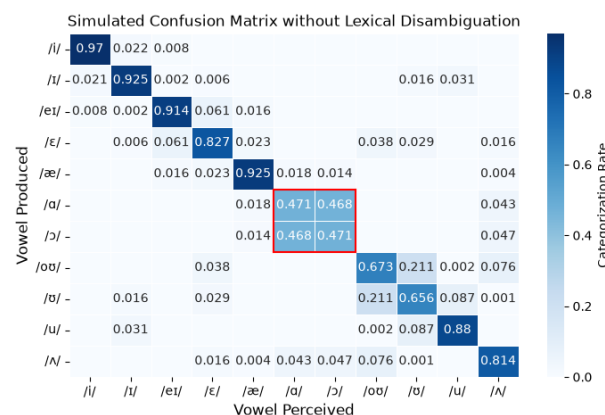
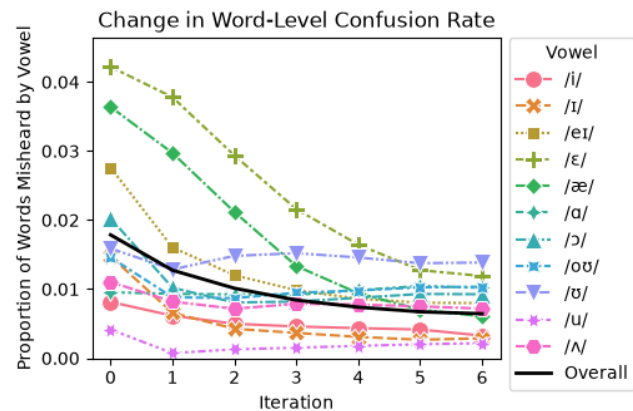
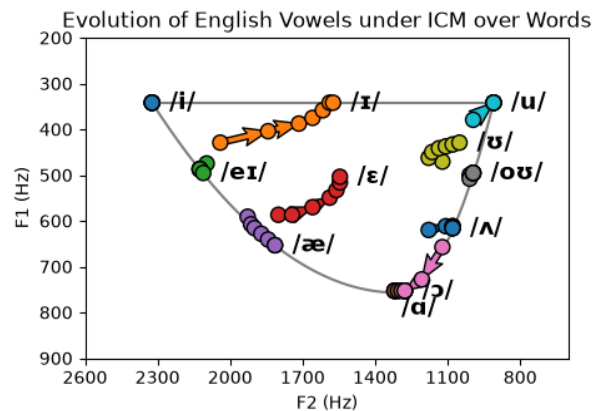
John McGahay (UCLA)

Accepted for an oral presentation at LSA 2027

The notion of pressure to maximize perceptual contrast between phonemic categories is thought to explain varied phonological phenomena including chain shifts (Martinet, 1955) and phonetic dispersion in vowels (Liljencrants & Lindblom, 1972). However, the common sound change phenomenon of phonemic merger (Labov, 1994), whereby the contrast between two historical phonemes is lost, presents a challenge: if languages are subject to contrast maximization pressures, why would a language ever lose a contrast? The *functional load hypothesis* (Gilliéron, 1918), which has found empirical support in statistical analysis of a cross-linguistic sound change data set (Wedel et al., 2013), posits that less important contrasts disintinguishing fewer minimal pairs of words are more likely to undergo merger. The current work makes a stronger claim: merger of contrasts with few minimal pairs directly results from contrast maximization pressures. Using a computational model called Iterated Confusion Minimization (McGahay, 2025), the current work shows that maximizing perceptual contrasts between words in US English predicts the common and well-known *cot-caught* merger (Labov et al., 2006) because it allows for reallocation of phonetic space to cue contrasts that distinguish a greater number of words.

Iterated Confusion Minimization (ICM) simulates the interaction of mathematically optimal listeners and speakers. Like existing Bayesian models of optimal listeners (Clayards et al., 2008; Feldman et al., 2009; Kleinschmidt, 2019), ICM models phonetic categories as probability distributions, allowing a mathematically precise measure of listener confusion based on distribution overlap. ICM carries out local optimization over this confusion measure and has been shown (McGahay, 2025) to predict sound change phenomena like chain shifts and to outperform past dispersion models (Schwartz et al., 1997; de Boer, 1999) in predicting vowel system typology.

Although ICM has previously been applied to individual vowel segments, the sensitivity of speech perception to words (Ganong, 1980) motivates us to generalize ICM beyond segments to words. The ICM simulation of this work starts with a system of stressed vowels with formants reported for US English by Hillenbrand et al. (1995). Each vowel is embedded in some context to make a word, e.g. vowels /ɑ/ and /ɔ/ can both occur in context /k_t/ to make *cot* and *caught*. The rate at which each word (vowel-context pair) occurs is taken from combining the CMU pronouncing dictionary with SUBTLEXus lexical frequencies. When this system is subjected to ICM over words (top left plot), /ɑ/ (brown) and /ɔ/ (pink) merge to a single phonetic value in the low back corner of the vowel space, resembling the *cot-caught* merger common in North American English. Even though simulated categorization of merged /ɑ/ and /ɔ/ based solely on vowel distributions is at chance (bottom left), simulated word-level confusion goes down over the course of the simulation (top right). This is because allowing the vowel to be disambiguated by knowledge of the lexicon results in highly accurate perception of both of these vowels (bottom right), which rarely occur in the same context (i.e. in minimal pairs) in English.



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